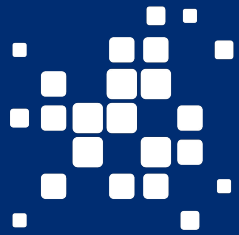
Genome comparisons with pan-k-mers

Presented by Katie



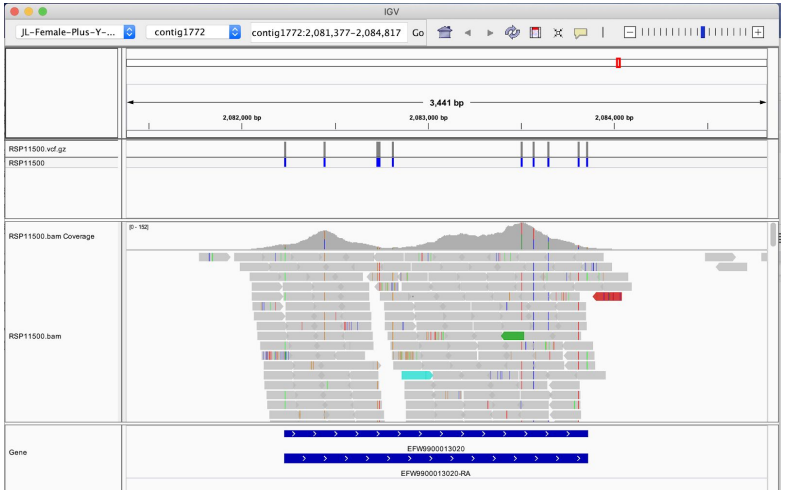
wellcome sanger institute

Outline

- **Example of a single pan-k-mer based tool**
  - Why pan-k-mers are useful
  - What can you do with a pan-k-mer bitmap
- Using pankmers to quickly identify introgressions
- Practical

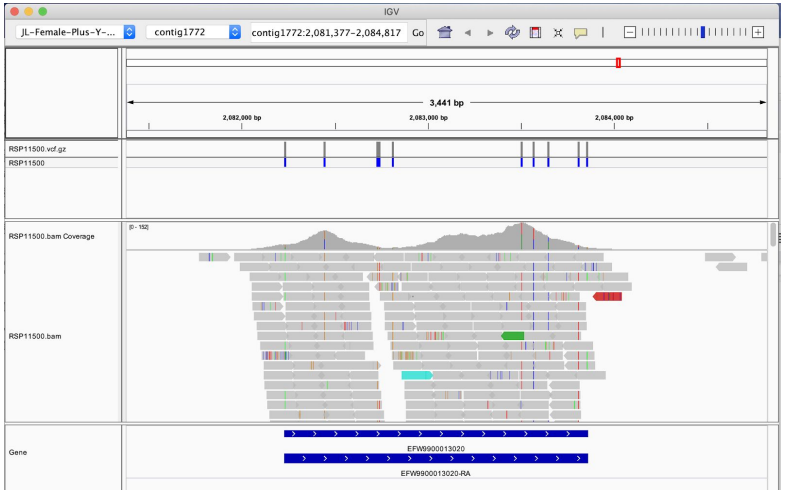
The challenge of visualizing and interpreting pangenomes

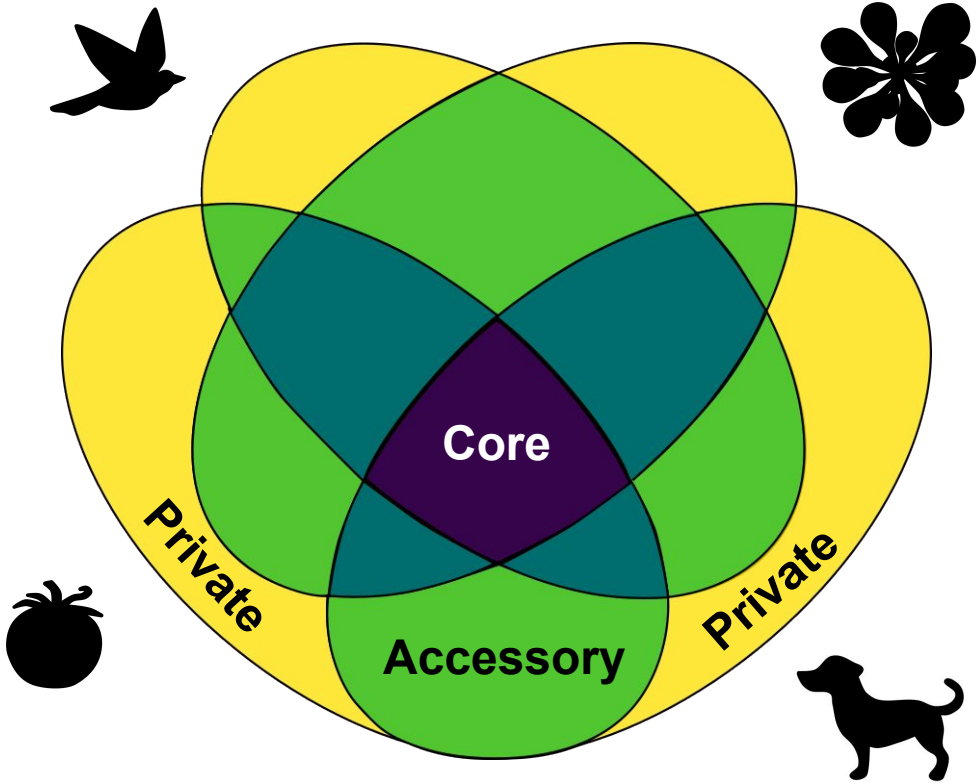
IGV screenshot



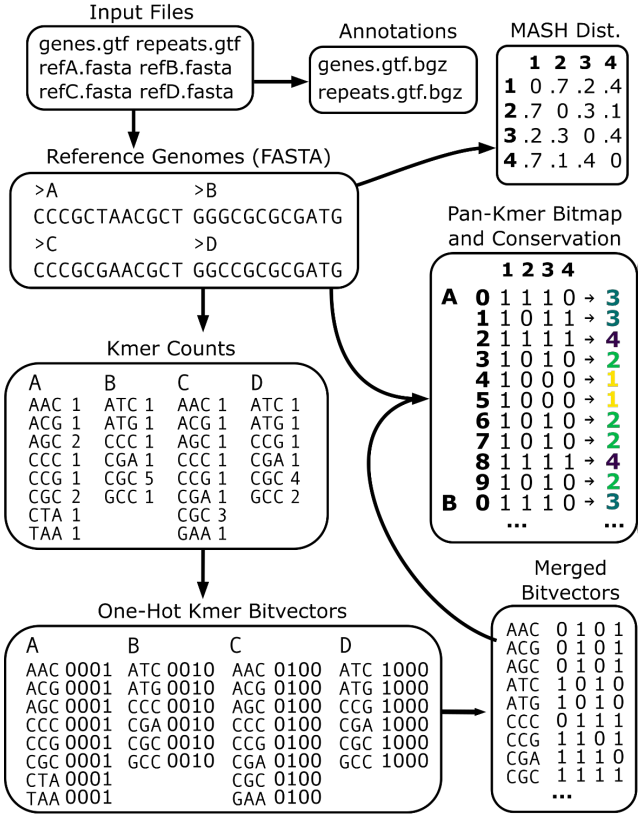
The challenge of visualizing and interpreting pangenomes

IGV screenshot





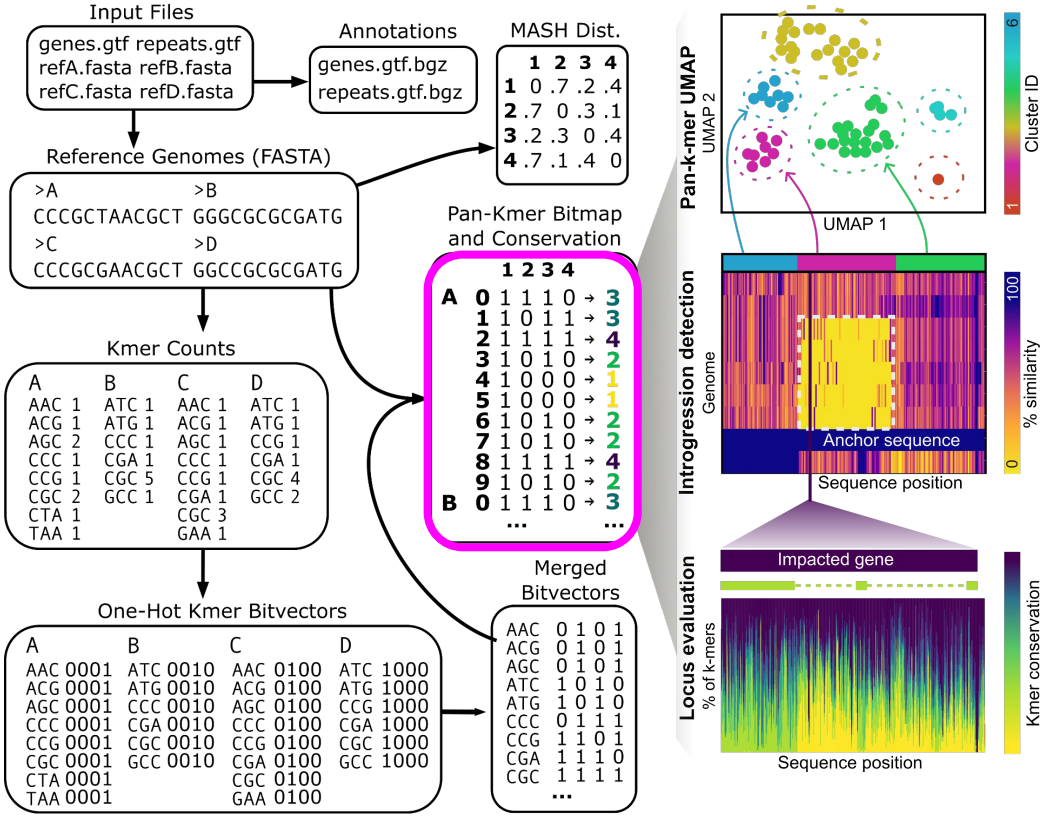
From sequence files to compressed pan-kmer bitmap to phenotypic insight



Sam Kovaka



From sequence files to compressed pan-kmer bitmap to phenotypic insight

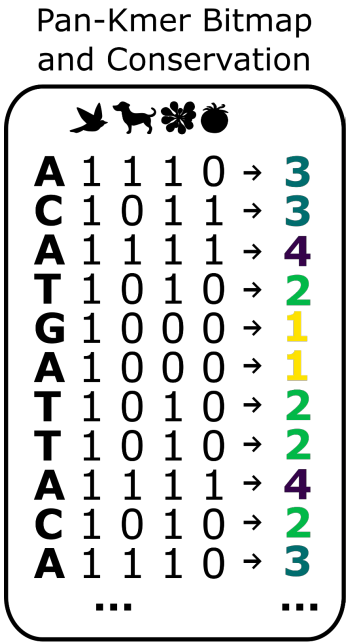




Sam Kovaka

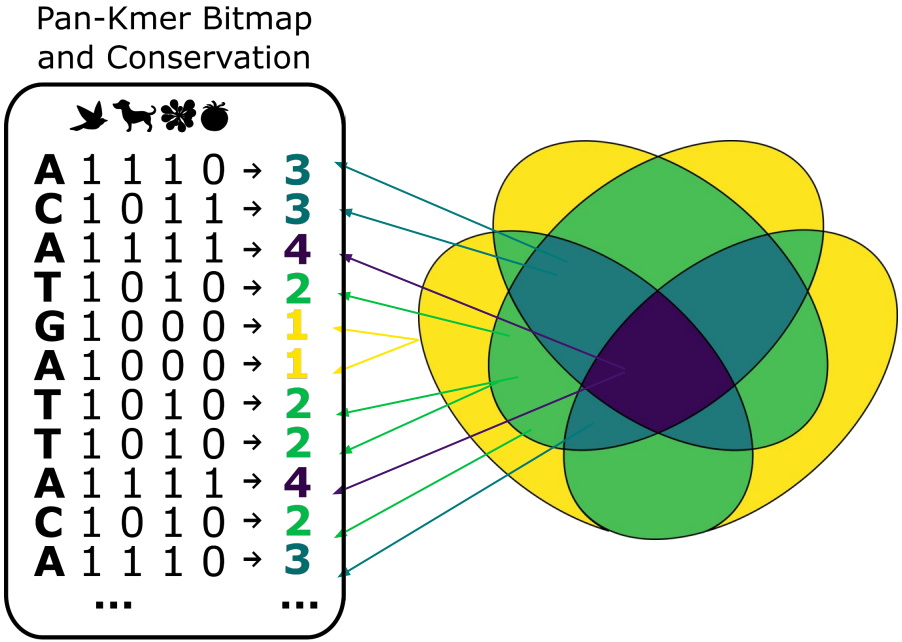
Pan-kmer bitmaps represent the presence/absence of every k-mer in a pan-genome

- The panagram bitmap includes the **presence/absence of every k-mer** in the pan-genome.
- This avoids multiple **whole genome pairwise comparisons** across a pan-genome.
- Pankmer bitmap **approximates an MSA**



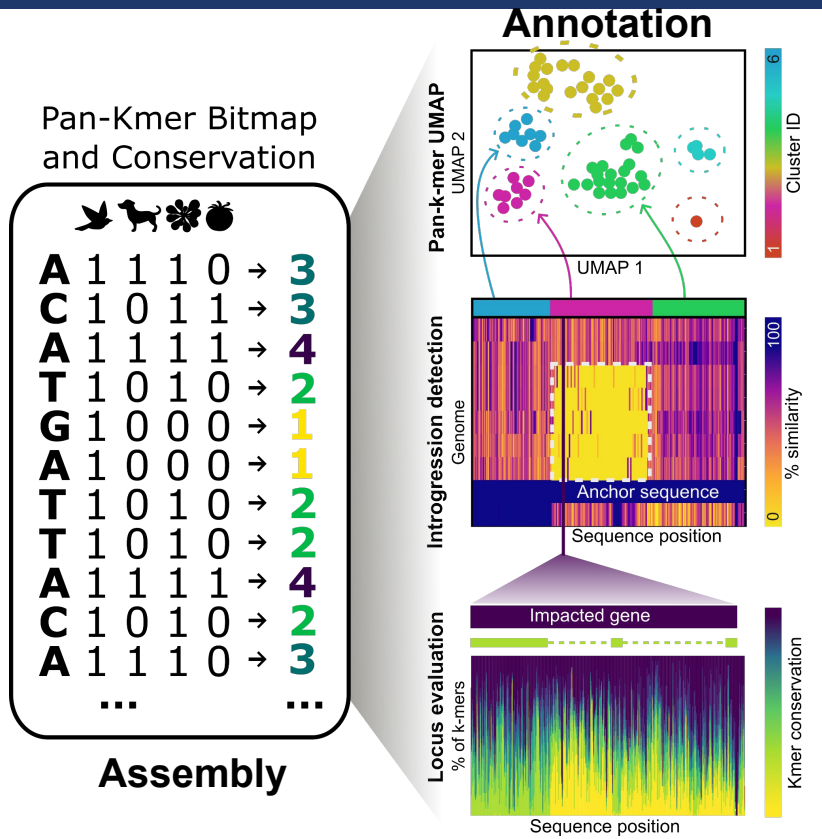
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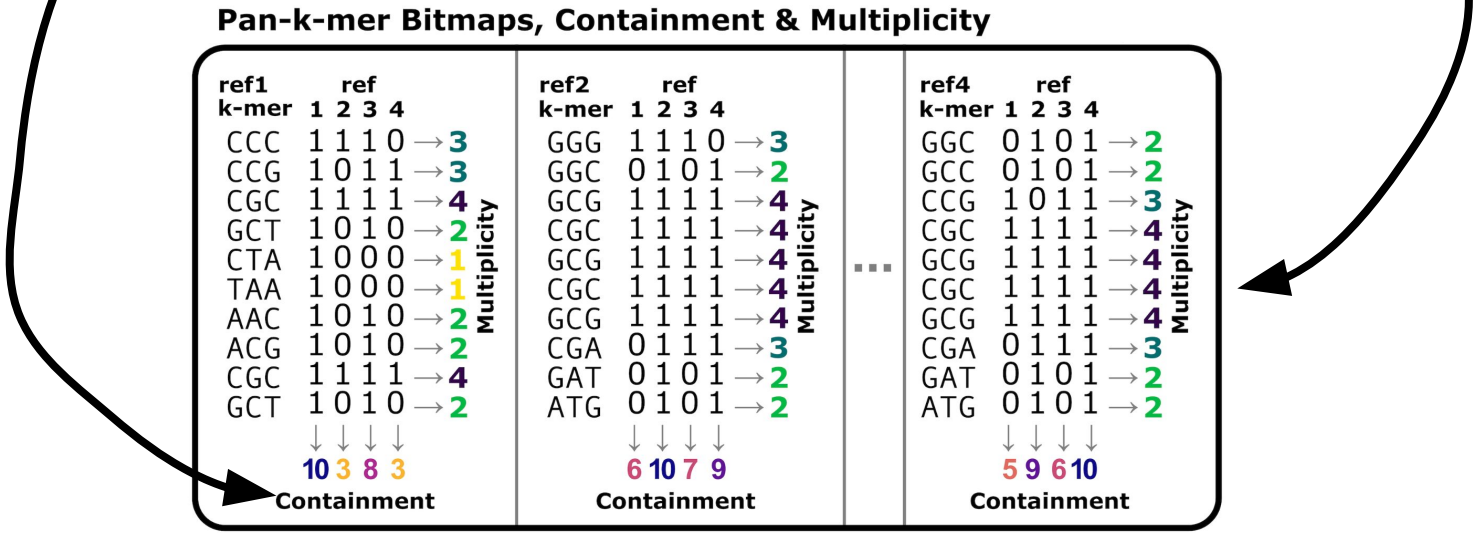
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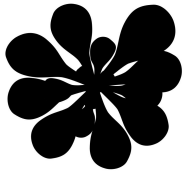
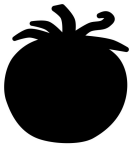
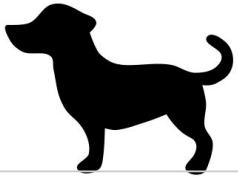


Pan-k-mer multiplicity and containment

- ***pan-k-mer multiplicity***: The number of genomes containing a given pan-k-mer, calculated as the *popcount* of its bitvector. This measures prevalence across genomes rather than copy number within a genome.
- ***pan-k-mer containment***: A measure of sequence identity between an interval of the anchor genome and a target genome.

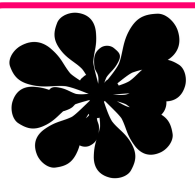
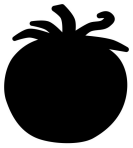
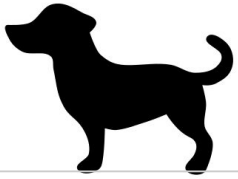


Pangenome analysis that scales with evolutionary distance and pangenome size

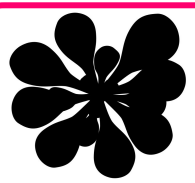
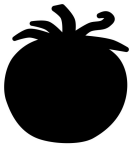
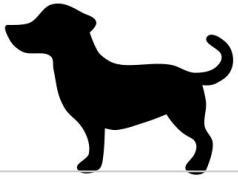

	A. thaliana	***Solanum***	**Canidae**	**"Zoonomia"**
	Species level	Genus level	Family level	Vertebrate level
# of genomes	160	24	50	605
Assembly file size	22GB	24GB	125GB	1002GB
Assembly type	Pacbio	Pacbio / ONT	Pacbio / ONT / Illumina	Pacbio / ONT / Illumina
Time to anchor individual	9 min.	10 min.	1hour 22 min.	14 hours, 39 min.
Time to run whole dataset	1 hour, 12 min.	47 min.	6 hours, 6 min.	8 days, 13 hours

Pangenome analysis that scales with evolutionary distance and pangenome size

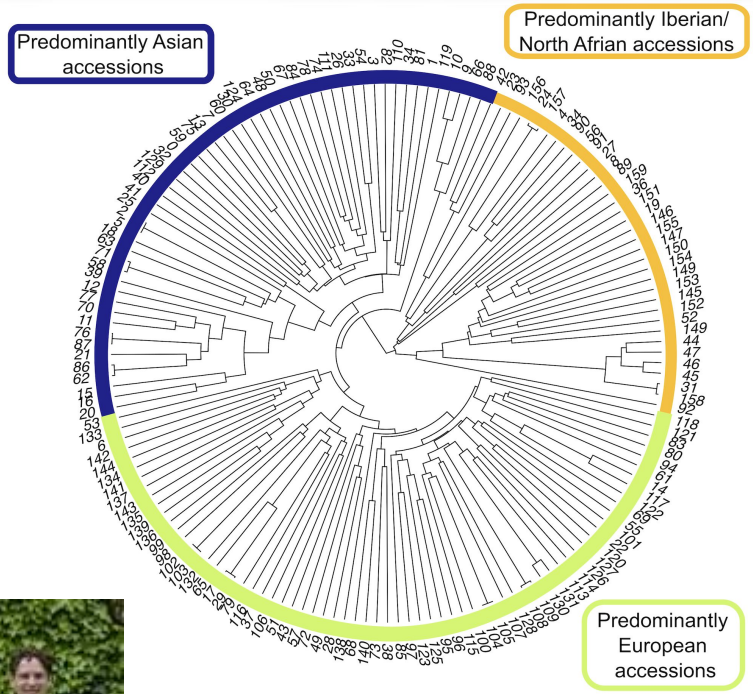

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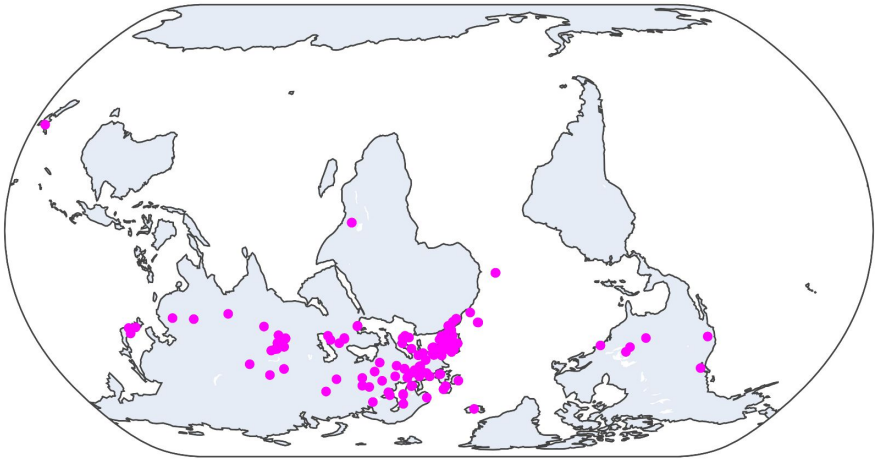
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Globally diverse *A. thaliana* genomes







Robin Burns

160 genomes took 1hr 12 minutes and 1 second with k=51

Assemblies from:
Wlodzimierz et al. Nature 2023, Lian et al. Nature Genetics 2024, Kang et al. Nat. Com. 2023

A. thaliana pangenome is annotated by Pan-k-mer multiplicity patterns

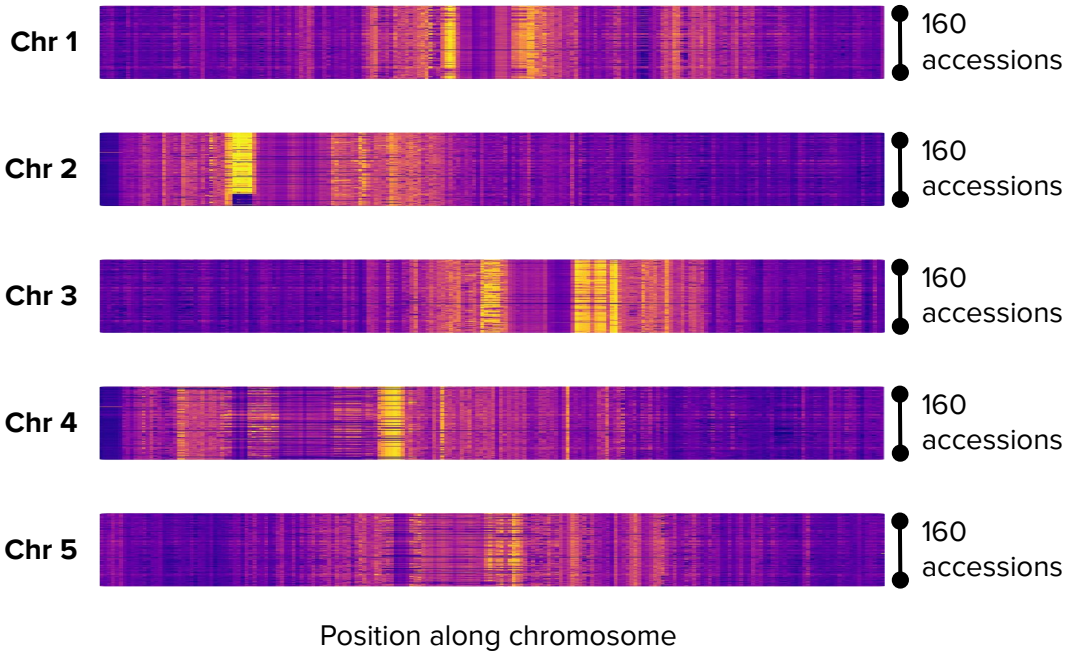


Dark purple = more similar to reference accession (Col-0)

Lighter = less similar to reference accession (Col-0)



Robin Burns



K=51

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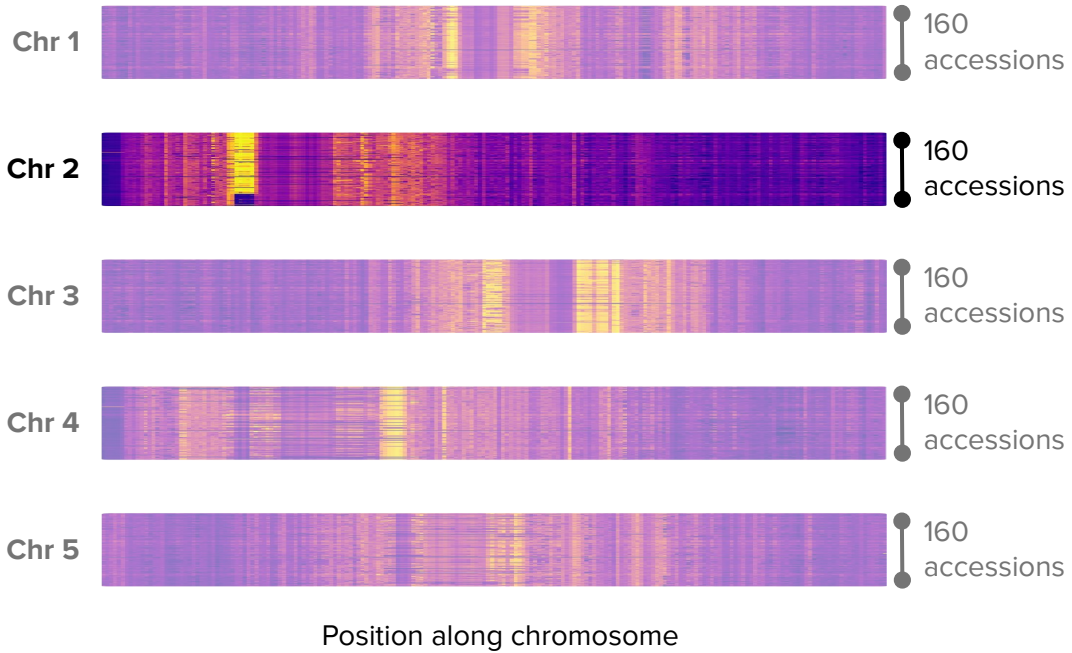


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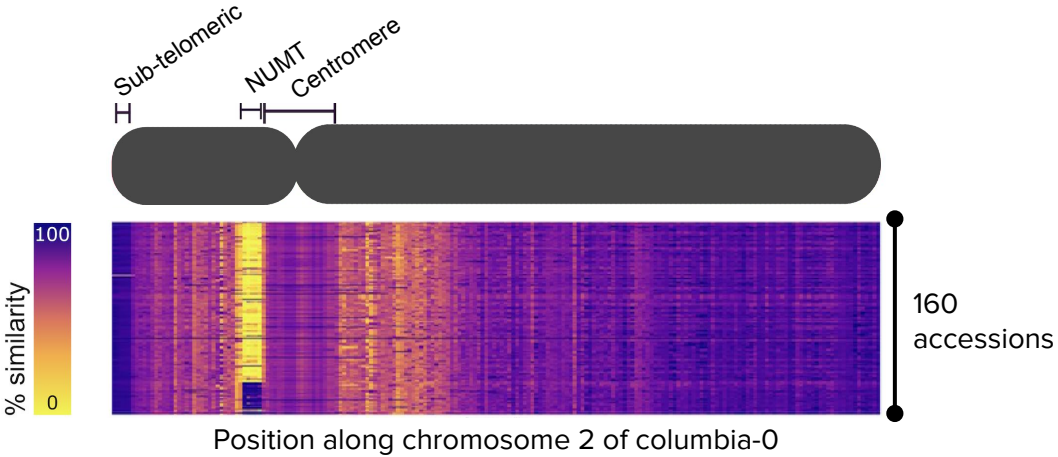


Robin Burns



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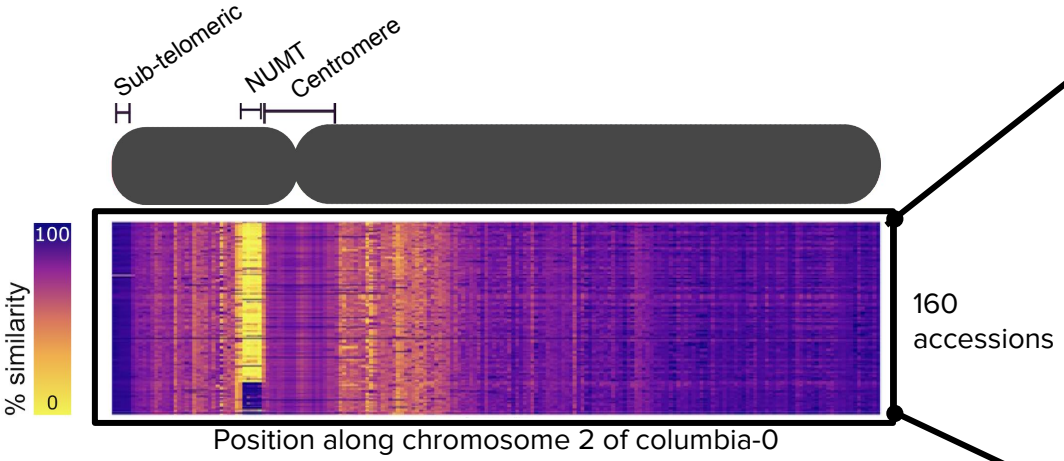
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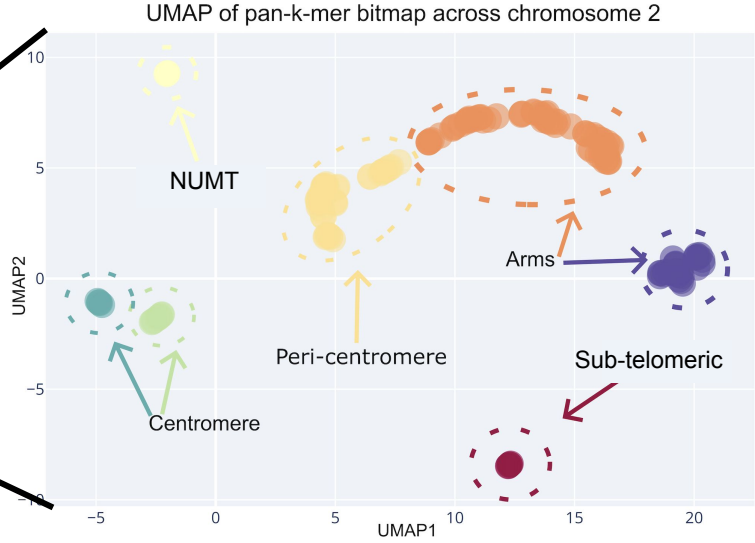
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Robin Burns

A. thaliana pangenome is annotated by Pan-k-mer multiplicity patterns





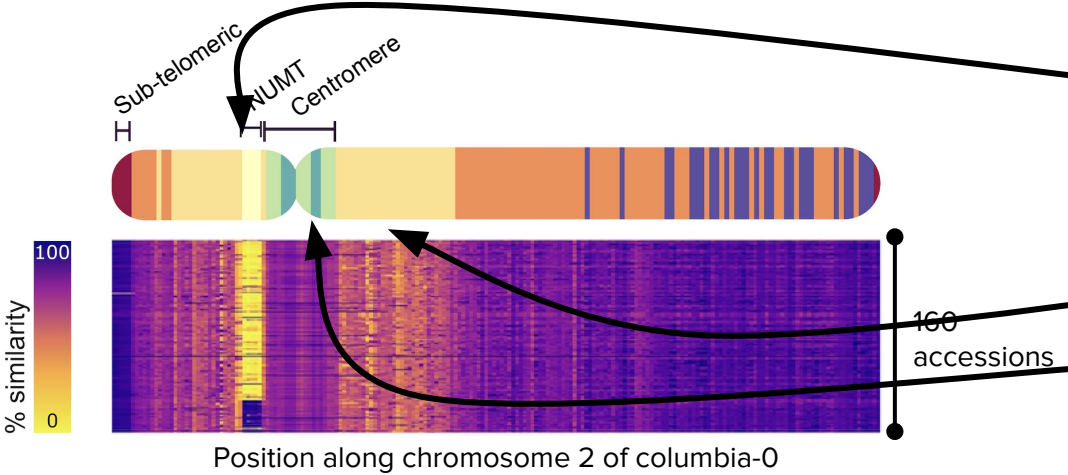
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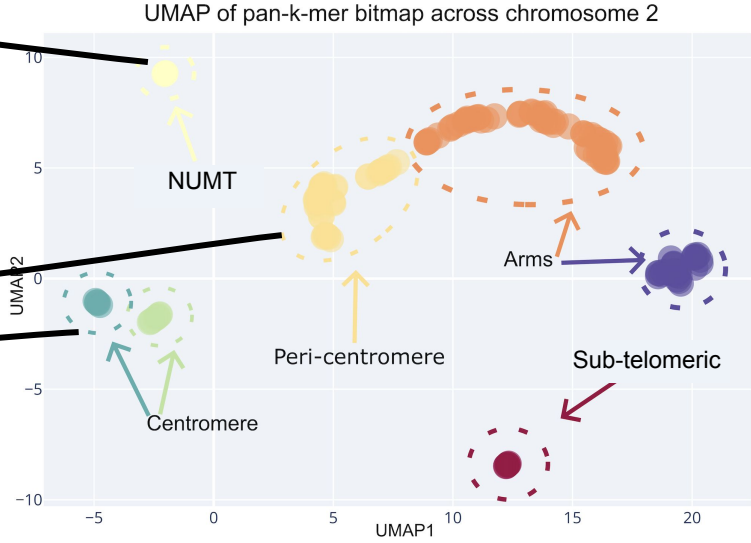
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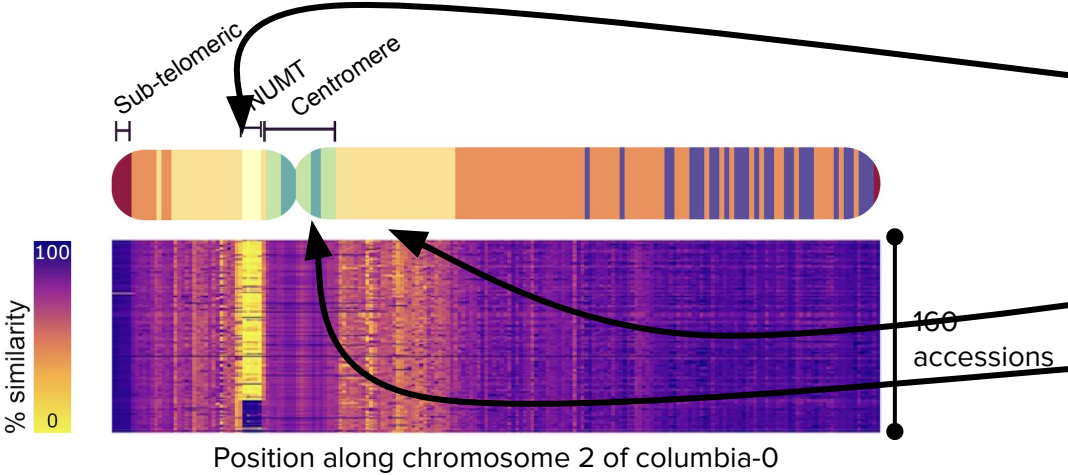
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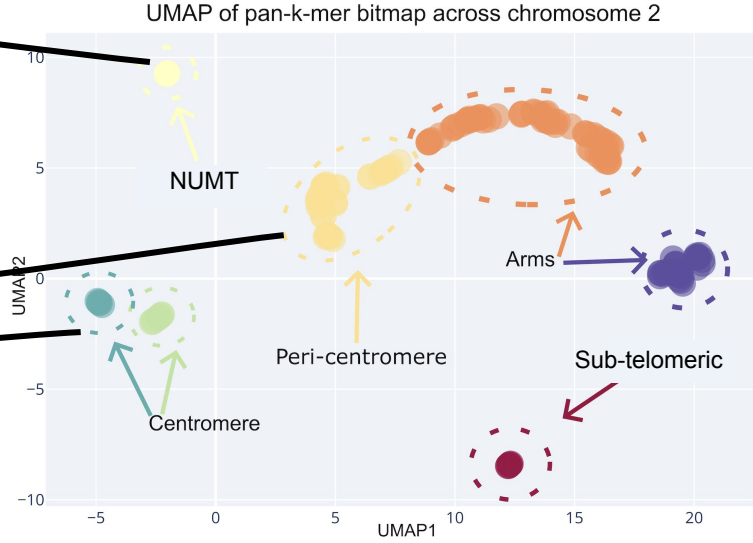
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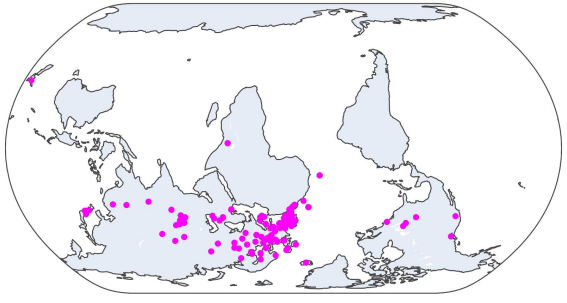
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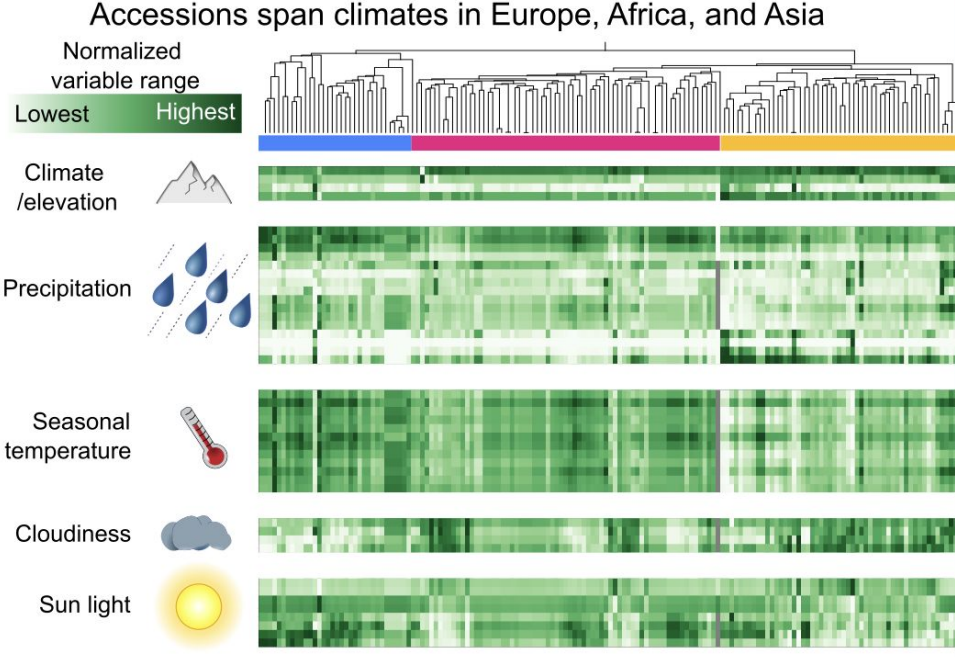


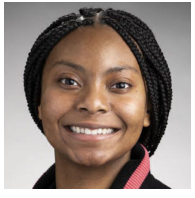
Robin Burns

Do haplotype patterns relate to phenotype?

A. thaliana pangenome is annotated by Pan-k-mer multiplicity patterns

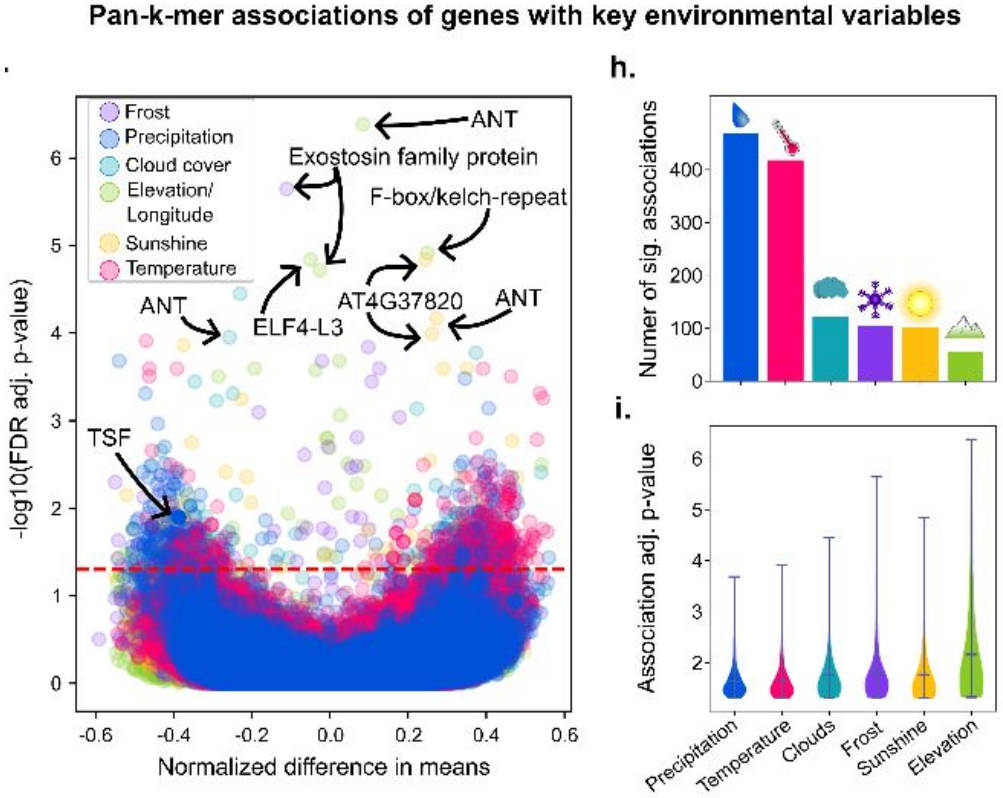






Nicole Brown

Hundreds of associations between haplotype structure and environmental conditions



Six newly assembled domesticated and mixed-breed genomes in Canis



Appa (M)
Sheepadoodle



Molly (F)
Greenland Village



Noodles (M)
Labrador retriever



Orfhlaith (F)
American Mutt



Pandan (M)
Doberman

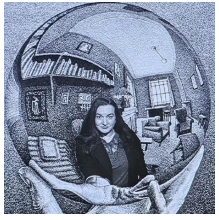


Zeke (M)
Bernedoodle

Elinor Karlsson



Delphine Lariviere



15 dog genomes ran in 2 hours, 1 minute

Six newly assembled domesticated and mixed-breed genomes in Canis



Appa (M)
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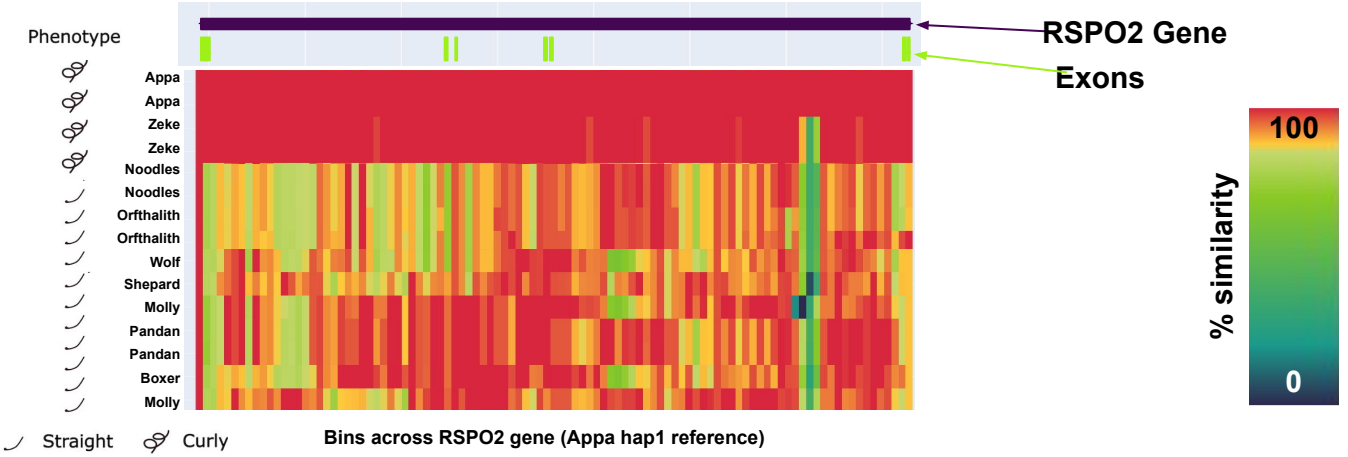
Orfhlaith (F)
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Doberman



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Cadieu *et al. Science*. 2009

Each row is a different dog, clustered by haplotype similarity

Six newly assembled domesticated and mixed-breed genomes in Canis



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Zeke (M)
Bernedoodle

\+ **Every published Canis genome**



Each row is a different dog, clustered by haplotype similarity

50 dog genomes ran in 6 hours, 6 minutes. Anchoring on just Appa_hap1 takes 1 hour 22 minutes

Six newly assembled domesticated and mixed-breed genomes in Canis





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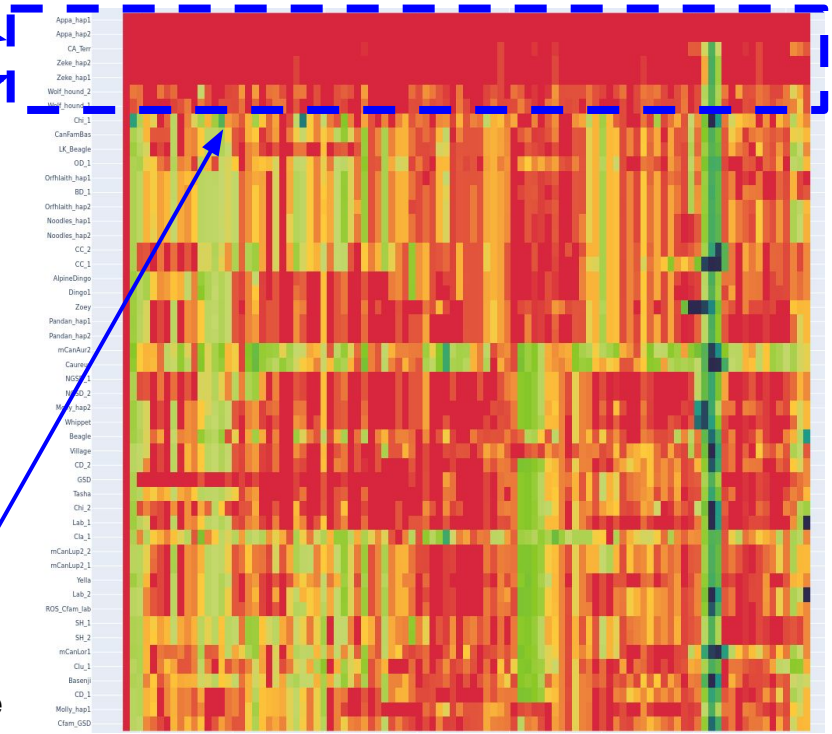
Pandan (M)
Doberman



Zeke (M)
Bernedoodle

\+ **Every published Canis genome**

Similar haplotype



Bins across RSPO2 gene (Appa hap1 reference



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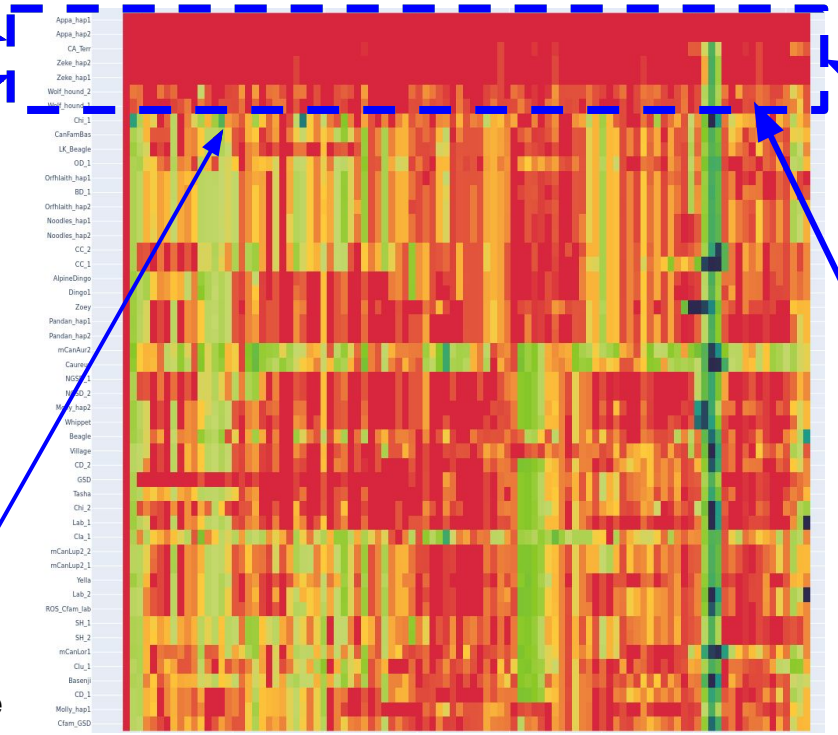
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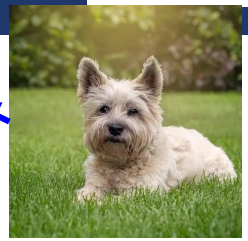


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+ Every published Canis genome

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% similarity

Bins across RSPO2 gene (Appa hap1 reference

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605 mammalian and avian genomes



Table 1 Aggregate statistics for the 600-way alignment

From: Progressive Cactus is a multiple-genome aligner for the thousand-genome era

Alignment	No. of genomes	Total bases	Instance-hours	Core-hours
Zoonomia	242	669 billion	68,166	1.9 million
B10K	363	400 billion	5,302	0.2 million
Combined	605	1.07 trillion	73,692	2.1 million

Article | Open access | Published: 11 November 2020

Progressive Cactus is a multiple-genome aligner for the thousand-genome era

Joel Armstrong, Glenn Hickey, Mark Diekhans, Ian T. Fiddes, Adam M. Novak, Alden Deran, Qi Fang, Duo Xie, Shaohong Feng, Josefin Stiller, Diane Genereux, Jeremy Johnson, Voichita Dana Marinescu, Jessica Alföldi, Robert S. Harris, Kerstin Lindblad-Toh, David Haussler, Elinor Karlsson, Erich D. Jarvis, Guojie Zhang & Benedict Paten

Nature **587**, 246–251 (2020) | Cite this article

605 mammalian and avian genomes

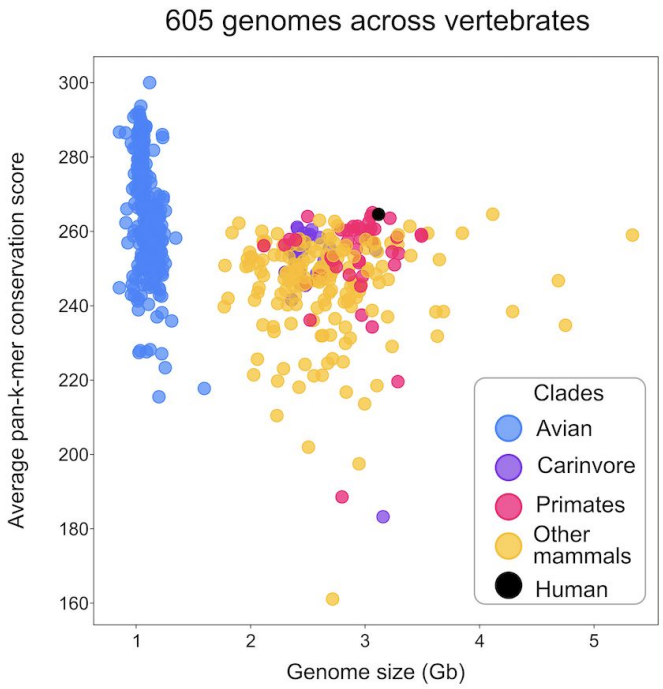


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Article | Open access | Published: 11 November 2020

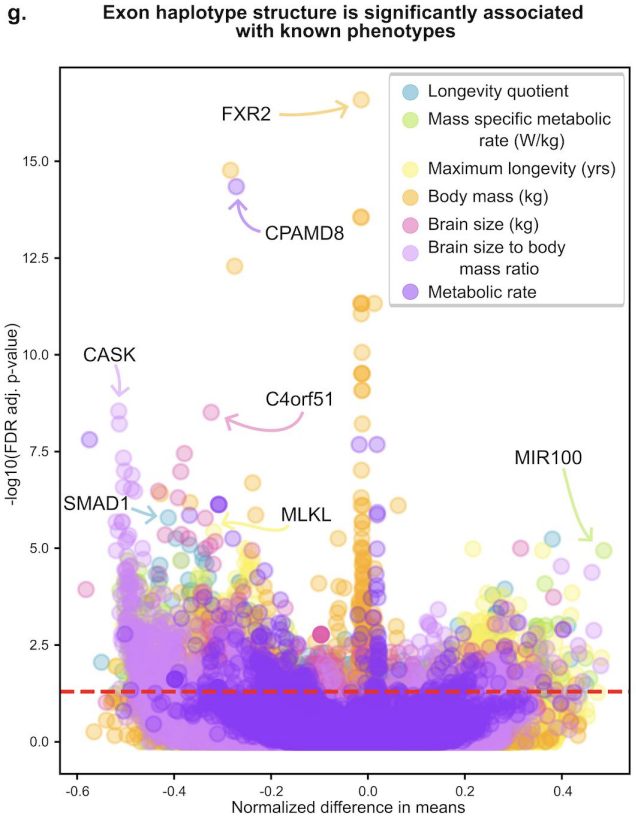
Progressive Cactus is a multiple-genome aligner for the thousand-genome era

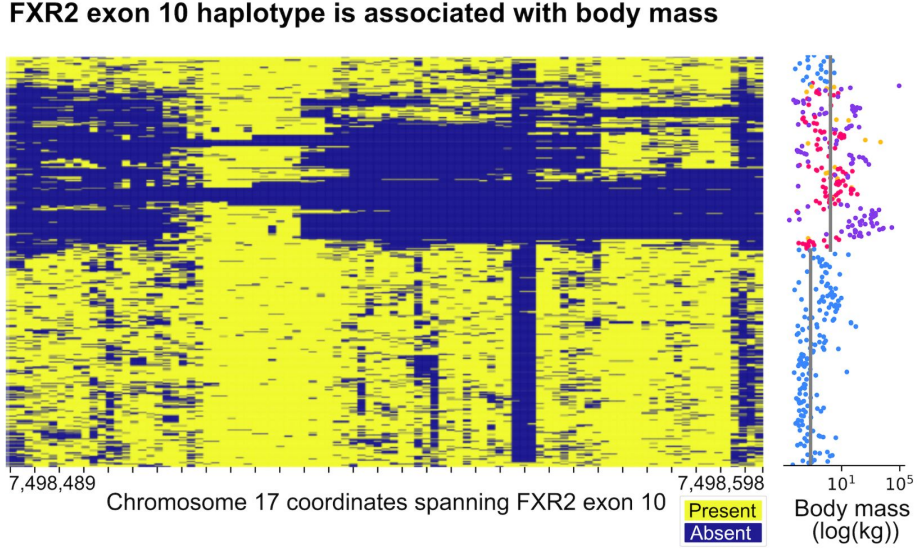
Joel Armstrong, Glenn Hickey, Mark Diekhans, Ian T. Fiddes, Adam M. Novak, Alden Deran, Qi Fang, Duo Xie, Shaohong Feng, Josefin Stiller, Diane Genereux, Jeremy Johnson, Voichita Dana Marinescu, Jessica Alföldi, Robert S. Harris, Kerstin Lindblad-Toh, David Haussler, Elinor Karlsson, Erich D. Jarvis, Guojie Zhang & Benedict Paten

Nature **587**, 246–251 (2020) | Cite this article

605 genomes ran in 8 days, 14 hours. **Processing just CHM13 takes 14 hours 39 minutes**

605 mammalian and avian genomes



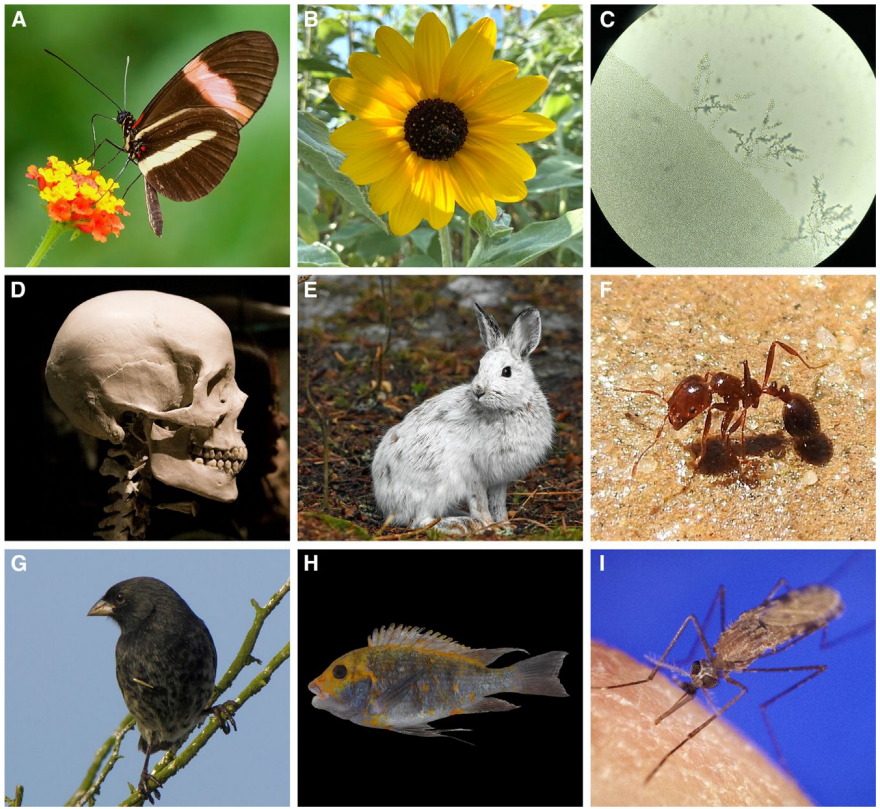


605 genomes ran in 8 days, 14 hours. **Processing just CHM13 takes 14 hours 39 minutes**

Outline

- Example of a single pan-k-mer based tool
  - Why pan-k-mers are useful
  - What can you do with a pan-k-mer bitmap
- **Using pankmers to quickly identify introgressions**
- Practical

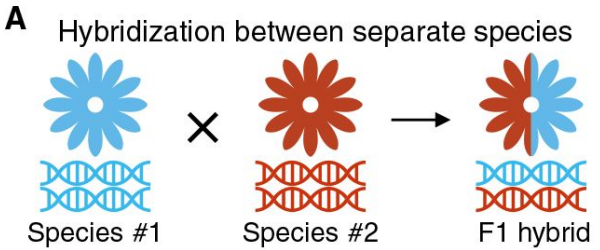
Introgressions are transfers of genetic material from one species to another

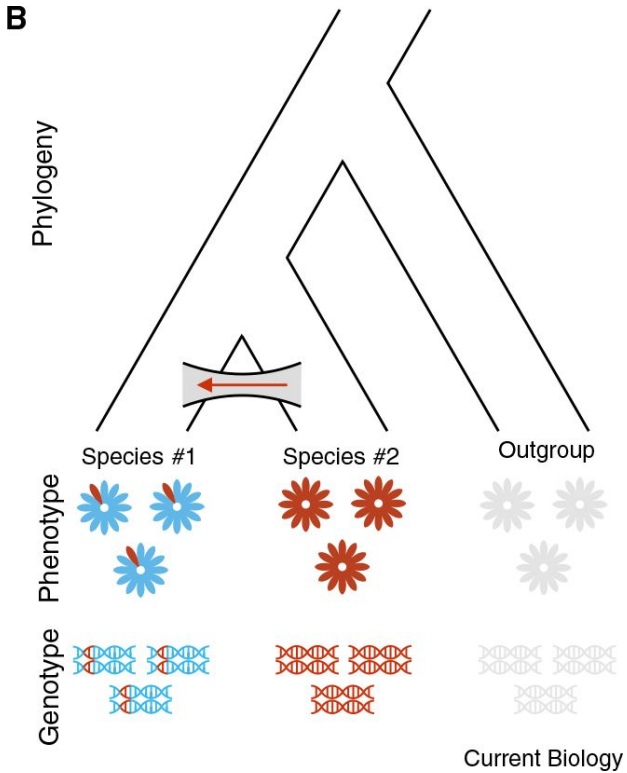


Introgression

Stepfanie M Aguillon [1,*], Tristram O Dodge [1], Gabriel A Preising [1], Molly Schumer [1]

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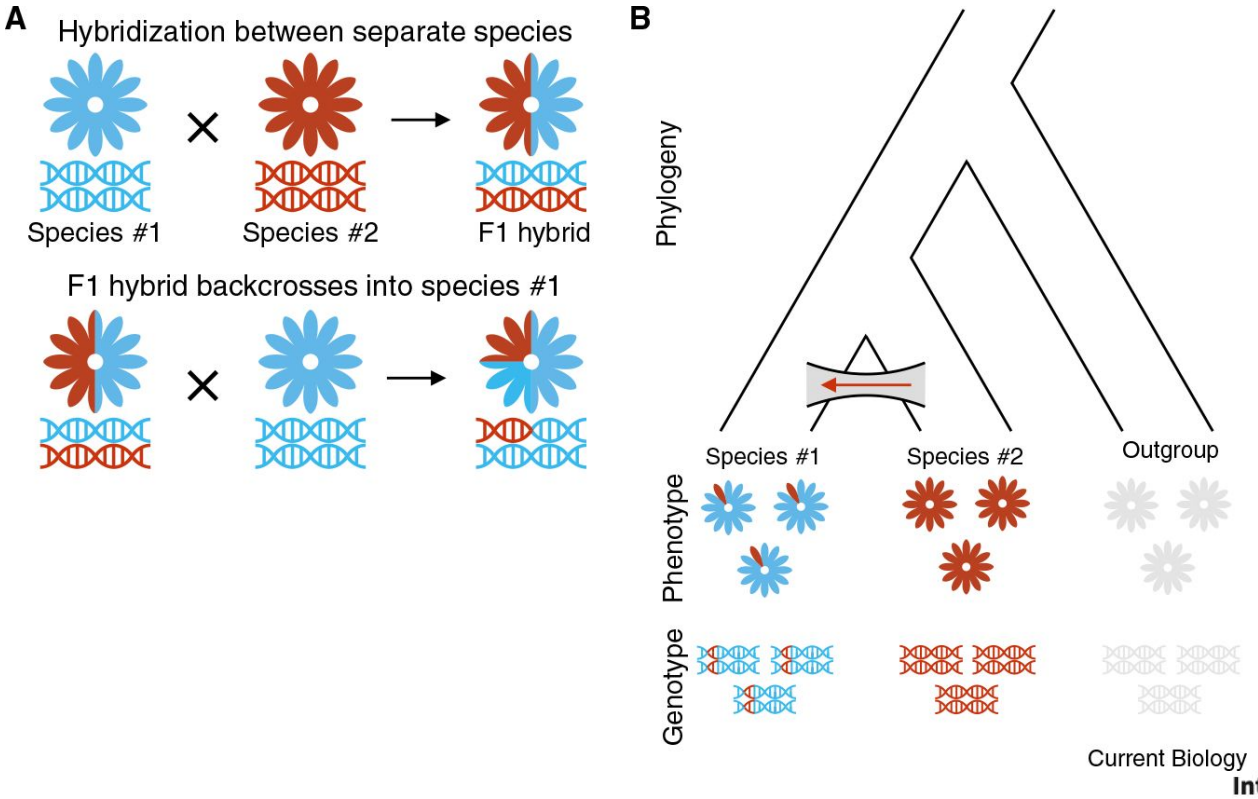


Current Biology

Introgression

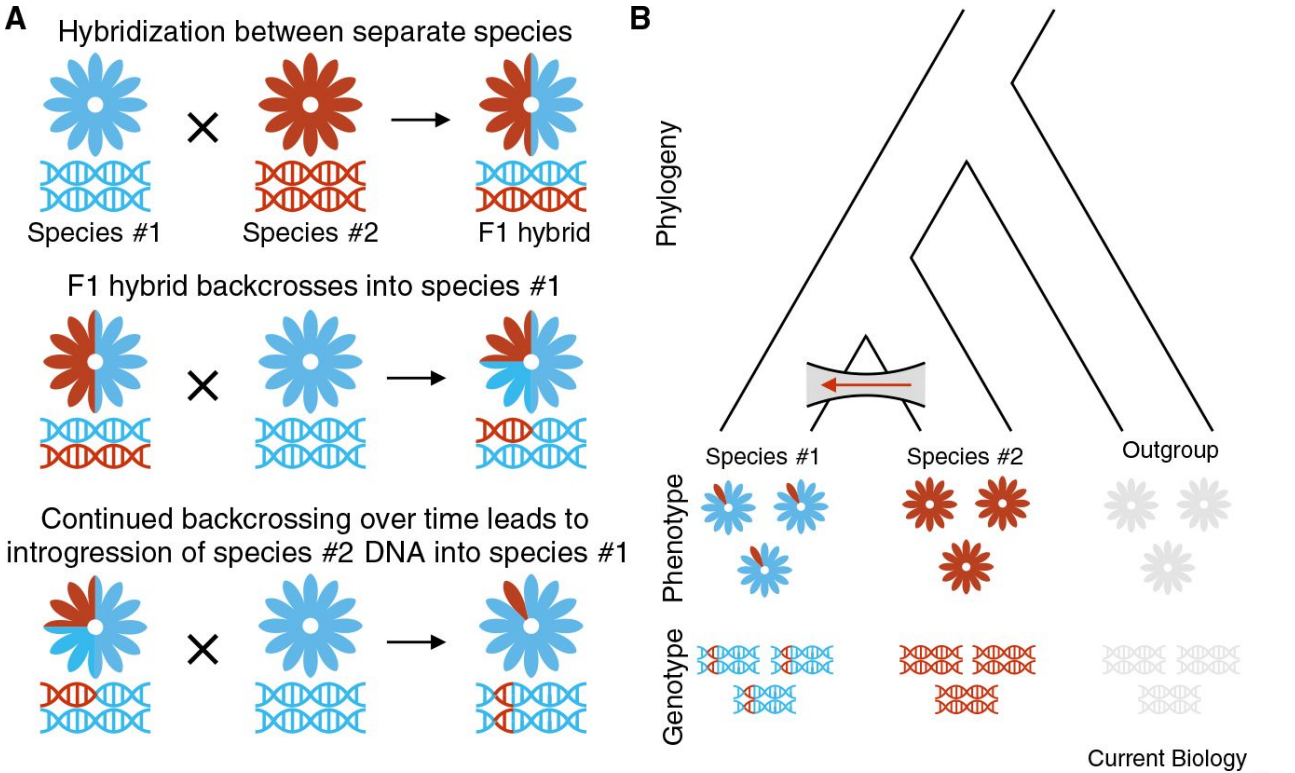
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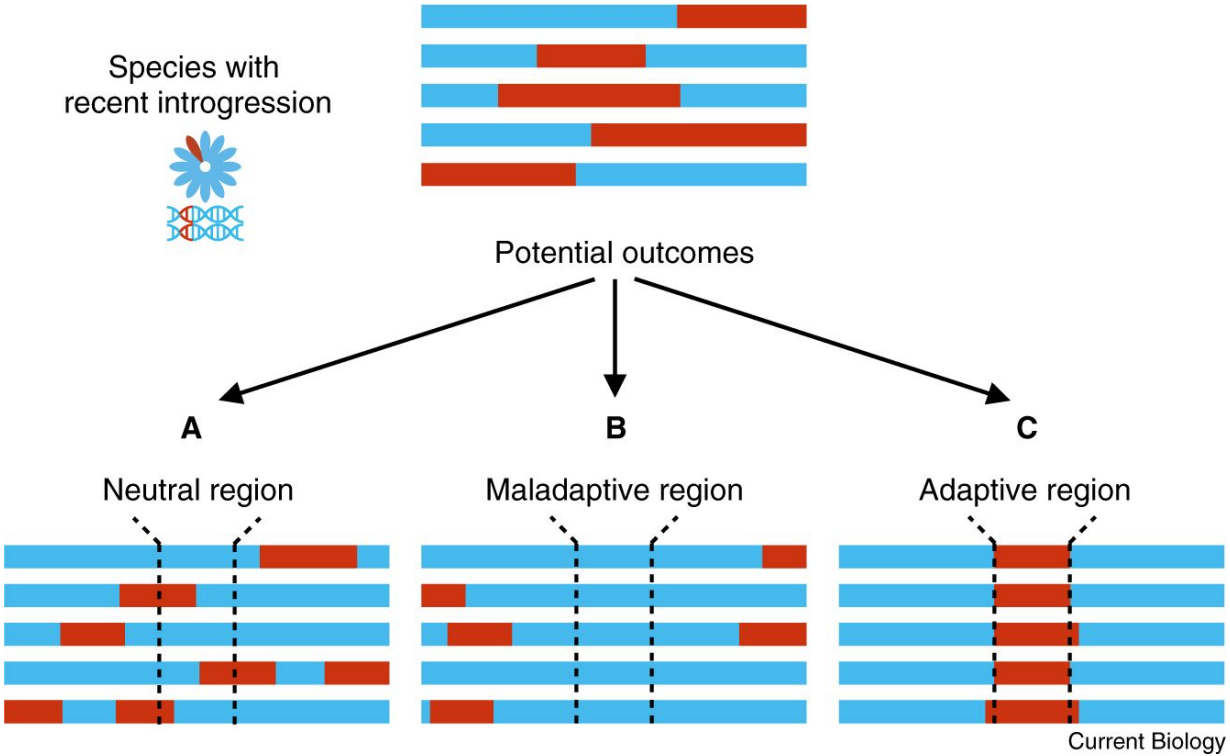
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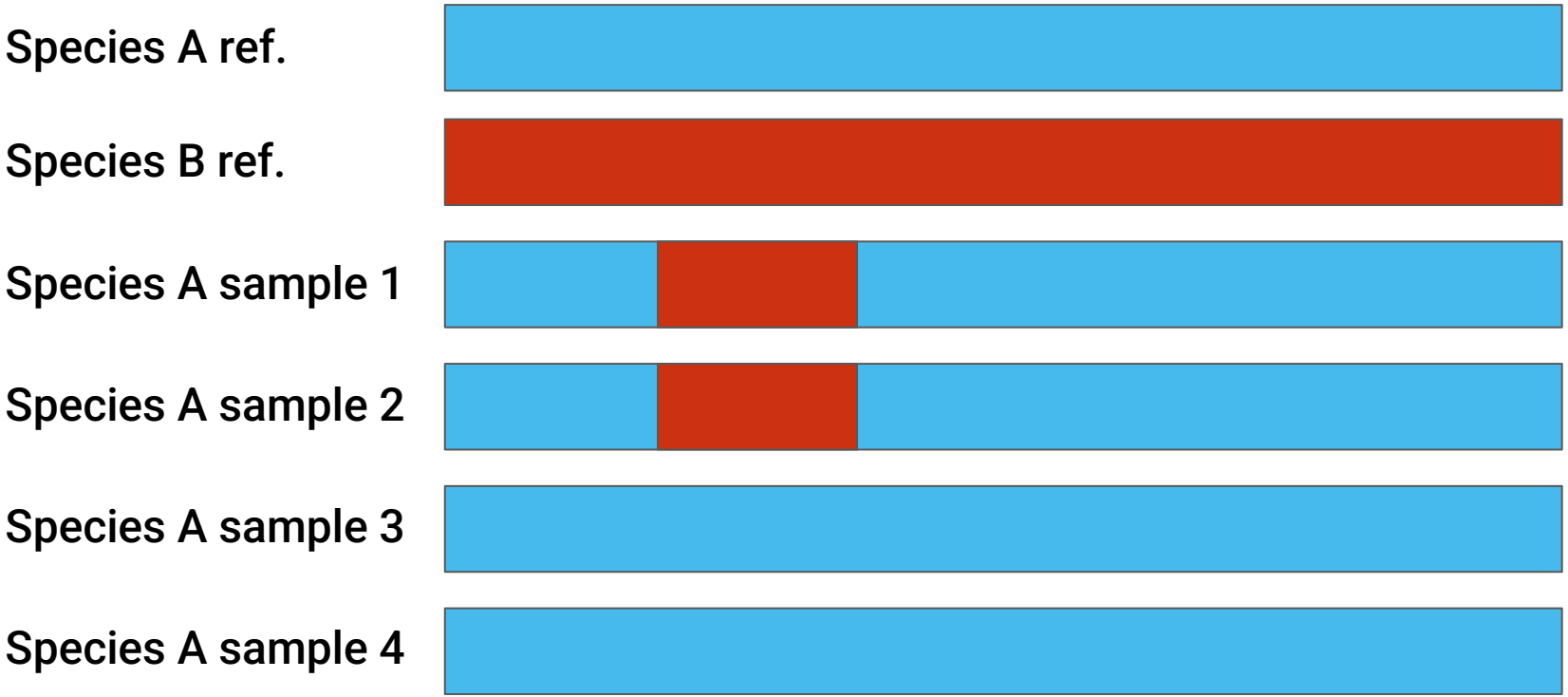
Species A ref.



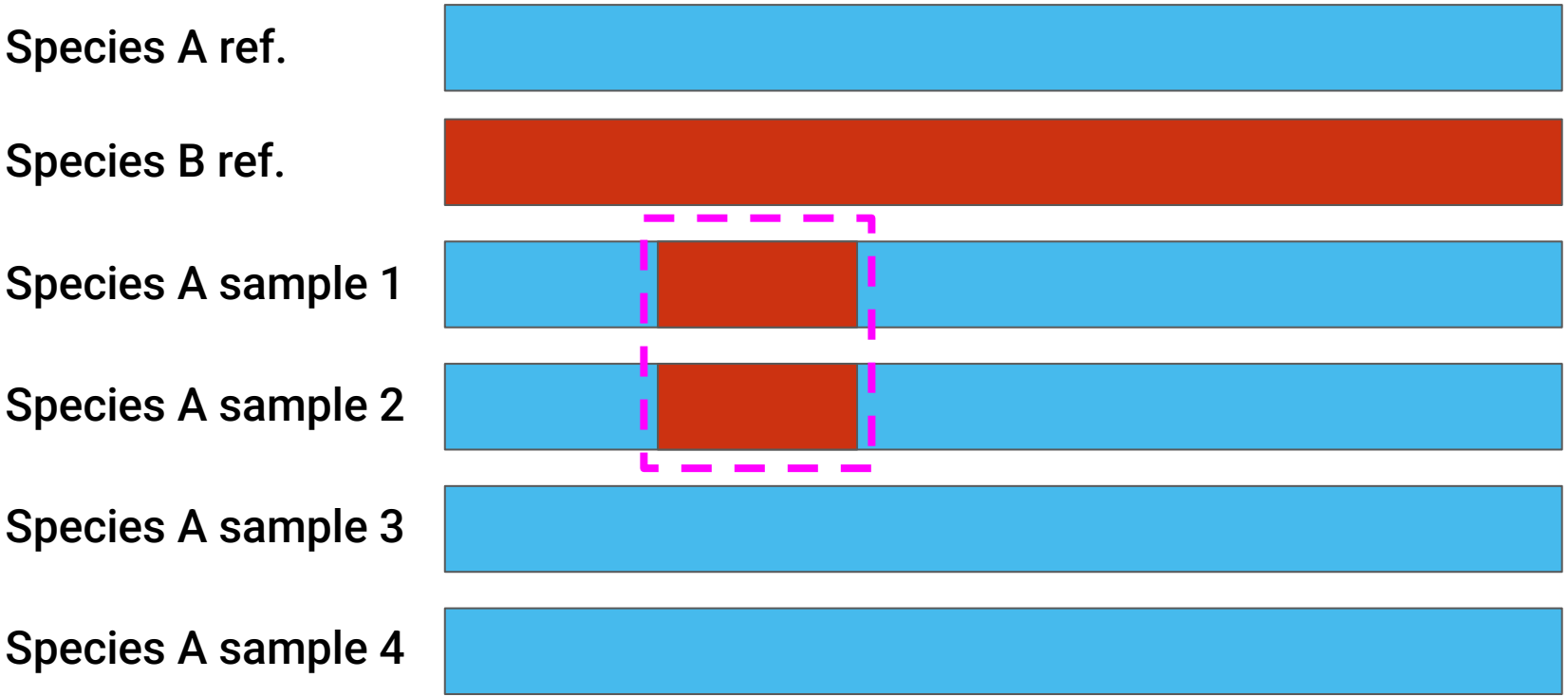
Species B ref.



Introgressions are transfers of genetic material from one species to another



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Outline

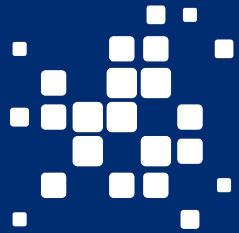
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Practical instructions

- We already ran Panagram for you on a set of *Solanum aethiopicum* genomes.
- There are suspected introgressions between *Solanum anguivi* (the wild relative) and *Solanum aethiopicum*.
- We will find them with pankmer bitmaps.

- Download the Panagram index
- Visualize each chromosome from the perspective of *S. aethiopicum* reference
- Visualize each chromosome from the perspective of *S. anguivi*
- Can you find an introgression? How do the pan-k-mer profiles compare between the introgressed regions and the rest of the genome?

Questions?



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